

Two-Phase PageRank with Travel-Time Self-Loops for Urban Traffic Distribution Estimation and Event-Driven Anomaly Detection

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Abstract

We present a modified PageRank algorithm for estimating link-level traffic distributions on urban road networks using sparse GPS trajectory data. The model introduces two key innovations: (1) *travel-time self-loops* that weight each link’s mass retention proportionally to its traversal time $\mu \cdot \ell_i / s_i$, and (2) a *two-phase block structure* separating active driving from local dispersal. Applied to the Salt Lake City road network (99,716 directed links, 467,911 GPS trajectories over 30 days), the model achieves a Top-100 Mean Relative Error (MRE) of 0.0312 and Pearson correlation $r = 0.997$ against smoothed ground truth.

We further introduce *demand-weighted transitions*, a single parameter γ that encodes average traffic demand into the transition matrix, enabling the damping factor to be raised from 0.80 to 0.96 (reducing ground-truth dependence from 20% to 4%) while *improving* overall accuracy. This is validated through temporal, day-of-week, and 5-fold cross-validation splits.

For anomaly detection, we demonstrate that the dwell-time parameter μ serves as an effective per-link control knob: reducing μ on disrupted links from 20 to 0.01 drops the crash-corridor prediction error from 179% to 11.5%, and game-day stadium-area error from 293% to 7.8%—all without using any event-day ground truth. Cross-validated Tuesday baseline analysis confirms $\mu = 20$ is optimal on normal days ($r = 0.9999$ between data gap and model error), establishing that deviations from this default reliably signal real-world disruptions.

1 Introduction

Accurate estimation of link-level traffic distributions is fundamental to transportation planning, real-time route guidance, and incident detection [11, 5]. Traditional approaches rely on fixed sensor networks (loop detectors, cameras), which provide high temporal resolution but limited spatial coverage. The proliferation of GPS-equipped mobile devices has created an alternative data source: sparse trajectory observations that cover the entire network but observe each link infrequently [3, 13].

The core challenge is converting sparse, noisy GPS traversal counts into a complete traffic distribution over all links. In a typical 15-minute observation window, only 1–3% of links in a metropolitan network have any GPS observations [4]. Directly using these counts produces a distribution dominated by zeros, with no information about unobserved links.

We address this through a three-stage pipeline:

1. **Graph-diffusion smoothing** propagates observed counts to neighbouring links using the road adjacency structure, filling gaps while preserving the spatial correlation of traffic patterns.

2. **Two-Phase PageRank with travel-time self-loops** computes a stationary distribution that balances observed data (teleportation) against network topology (transition matrix), producing physically realistic traffic estimates.
3. **Demand-weighted transitions** encode average traffic demand into the matrix, enabling high damping factors ($d > 0.90$) that reduce dependence on per-timeframe ground truth.

The remainder of this paper is organised as follows. Section 2 describes the data pipeline. Section 3 presents the two-phase model. Section 4 introduces demand-weighted transitions. Section 5 reports accuracy results. Section 6 demonstrates event detection and prediction. Section 7 analyses the role of μ . Section 8 concludes.

2 Data Pipeline

2.1 Study Area and Network

The study area is the greater Salt Lake City metropolitan region, represented as a directed graph $G = (V, E)$ with $|E| = N = 99,716$ directed links and average out-degree 2.87. The network is derived from OpenStreetMap via SUMO, with each link carrying attributes: length ℓ_i (metres), free-flow speed s_i (m/s), and lane count. Link lengths range from 0.59 m to 11,577 m (mean 142 m), and speeds from 2.2 to 33.5 m/s.

2.2 GPS Trajectory Data

We use 467,911 GPS trajectories recorded between August 31 and September 30, 2018. Each trajectory consists of timestamped (*latitude, longitude*) points from mobile devices. The raw data is processed through:

1. **Cleaning:** Speed filtering ($v > 2$ m/s), 5-minute gap splitting into separate trips.
2. **Map matching:** Hidden Markov Model-based matching [7] using the Leuven map-matching library [6], producing sequences of traversed link IDs with entry timestamps.
3. **Temporal binning:** Counting traversals per link per 15-minute window, yielding a $2,880 \times 63,118$ sparse count matrix (2.06% density).

2.3 The Sparsity Problem

In any 15-minute window, only 1–3% of network links have GPS observations. The raw count vector c_t for timeframe t has $c_t[i] = 0$ for $\sim 97\%$ of links. Directly using $c_t/\|c_t\|_1$ as a traffic distribution assigns zero probability to most links, making the Mean Relative Error (MRE) undefined for unobserved links.

2.4 Graph-Diffusion Smoothing

We address sparsity through a one-step graph-diffusion filter [9]:

$$c_{\text{smooth}} = [(1 - \gamma) I + \gamma P_{\text{adj}}] \tilde{c} \quad (1)$$

where $\tilde{c} = c + \alpha \cdot \mathbf{1}$ is the Laplace-smoothed count vector ($\alpha = 0.1$), P_{adj} is the row-stochastic adjacency matrix, and $\gamma \in [0, 1]$ is the diffusion strength.

Spectral interpretation. P_{adj} has eigenvalues $\lambda \in [-1, 1]$. The filter $(1 - \gamma)I + \gamma P_{\text{adj}}$ has eigenvalues $(1 - \gamma) + \gamma\lambda$. The DC component ($\lambda = 1$) is preserved exactly; high-frequency noise ($\lambda \approx -1$) is attenuated by factor $|1 - 2\gamma|$. At $\gamma = 0.26$, attenuation is 48%.

Cross-validation. We determine γ via split-half cross-validation (49 random timeframes, seed 42), measuring both predictive MSE and downstream PageRank MRE. The optimal $\gamma = 0.26$ minimises both metrics simultaneously, retaining 69% of the original variance. After smoothing, the matrix becomes $2,880 \times 99,716$ with 100% density—every link has a positive probability estimate.

3 Two-Phase PageRank Model

3.1 Standard PageRank

The classical PageRank [8, 1] computes the stationary distribution of a random walk on a graph:

$$v^{(k+1)} = d \cdot P^\top v^{(k)} + (1 - d) \cdot E \quad (2)$$

where d is the damping factor, P is a row-stochastic transition matrix, and E is a teleportation vector (uniform in the original formulation, or empirical in our case).

3.2 Travel-Time Self-Loops

In a road network, vehicles spend time *on* each link proportional to its traversal time ℓ_i/s_i . Standard PageRank treats all transitions as instantaneous, assigning equal weight to short connectors and long highways. We introduce self-loops that model dwell time:

$$\boxed{sw_i = \mu \cdot \frac{\ell_i}{s_i}} \quad (3)$$

where $\mu > 0$ is the dwell-time scaling parameter. The transition probability from link i to successor j becomes:

$$P[i, j] = \frac{w_{ij}}{sw_i + \sum_k w_{ik}}, \quad P[i, i] = \frac{sw_i}{sw_i + \sum_k w_{ik}} \quad (4)$$

For a typical highway link ($\ell = 500$ m, $s = 25$ m/s, 3 successors):

$$sw = 20 \times 500/25 = 400, \quad P[i, i] = \frac{400}{400 + 3} = 99.3\%$$

After k iterations, the retained mass is $P[i, i]^k$. This naturally allocates more PageRank mass to links with longer traversal times—which are precisely the high-volume corridors that carry the most traffic.

3.3 Two-Phase Block Structure

We decompose the random walk into two phases:

- **Up phase** (active driving): follows the directed adjacency with travel-time self-loops.
- **Down phase** (local dispersal): vehicles that have “arrived” disperse locally.

The $2N \times 2N$ transition matrix is:

$$M_{2N} = \begin{bmatrix} (1 - \beta) P_{\text{up}} & \beta I_N \\ \mathbf{0} & P_{\text{down}} \end{bmatrix} \quad (5)$$

where $\beta \in (0, 1)$ is the phase-drop rate. At each step, a fraction β of up-phase mass transitions to the down phase. The optimal $\beta = 0.9$ implies the average trip has approximately $1/\beta \approx 1.1$ forward hops before dispersal.

3.4 Power Iteration

The full update rule is:

$$v^{(k+1)} = d \cdot M_{2N}^\top v^{(k)} + (1 - d) \cdot E_{2N} \quad (6)$$

where $E_{2N} = [E_N; \mathbf{0}]$ injects the smoothed ground truth into the up phase only, and $d = 0.80$ is the damping factor. Convergence is measured by $\|v^{(k+1)} - v^{(k)}\|_1 < 10^{-6}$ (typically 37–38 iterations).

The final distribution is obtained by collapsing phases:

$$v_{\text{final}} = \frac{v_{\text{up}} + v_{\text{down}}}{\|v_{\text{up}} + v_{\text{down}}\|_1} \quad (7)$$

4 Demand-Weighted Transitions

4.1 Motivation

At $d = 0.80$, 20% of the prediction comes from ground truth. To reduce this dependence, we increase d —but the topology-only matrix does not encode traffic demand patterns, so accuracy degrades rapidly (Table 3).

A grid search over 576 parameter combinations $(\mu, \alpha_s, \alpha_l, \beta)$ found *zero feasible solutions* for $d > 0.90$ with MRE ≤ 0.05 .

4.2 The γ Parameter

We introduce a single parameter γ that weights each transition by the destination link’s average demand:

$$P(i \rightarrow j) \propto w_{ij} \cdot \text{demand}[j]^\gamma \quad (8)$$

where $\text{demand}[j] = \frac{1}{T} \sum_{t=1}^T E_N^{(t)}[j]$ is the time-averaged popularity of link j across a training set of T timeframes. This teaches the matrix that high-traffic links (e.g., I-15) should attract more flow than residential streets.

4.3 Validation

We validate through three independent train/test splits:

1. **Temporal:** Train on Sept 1–15, test on Sept 16–30.
2. **Day-of-week:** Train on Mon/Wed/Fri, test on Tue/Thu/Sat/Sun.
3. **5-fold random CV:** 80% train, 20% test, repeated 5 times.

Results (Table 1) confirm the improvement generalises: at $d = 0.96$, $\gamma = 0.20$, the average Overall MRE across all splits is **0.0385**—better than the original model at $d = 0.80$ (0.0466), while using only 4% ground truth.

Table 1: Overall MRE across three validation strategies.

Configuration	Temporal	DoW	5-Fold	Avg
$d=0.80, \gamma=0$	0.0478	0.0457	0.0464	0.0466
$d=0.91, \gamma=0.10$	0.0472	0.0452	0.0459	0.0461
$d=0.94, \gamma=0.15$	0.0439	0.0420	0.0427	0.0429
$d=0.96, \gamma=0.20$	0.0395	0.0377	0.0383	0.0385

Table 2: Evaluation metrics (49 timeframes, $\mu=20$, $\beta=0.9$, $d=0.80$).

Metric	Mean $\pm$ Std	Range
Top-100 MRE	0.0312 ± 0.0122	0.020–0.048
Overall MRE	0.0464 ± 0.0056	0.041–0.057
Pearson r	0.9969 ± 0.0014	0.995–0.998
Rank overlap /100	94.4 ± 4.5	86–99

5 Results

5.1 Evaluation Protocol

We evaluate on 49 random timeframes (seed 42) held out from parameter tuning. The ground truth is the smoothed popularity vector E_N for each timeframe. The prediction is v_{final} from the two-phase PageRank. The primary metric is Mean Relative Error:

$$\text{MRE} = \frac{1}{|S|} \sum_{i \in S} \frac{|E_N[i] - v_{\text{final}}[i]|}{E_N[i] + \epsilon} \quad (9)$$

computed over all links (Overall) or the top-100 busiest links (Top-100).

5.2 Main Results

Table 2 shows the model achieves Top-100 MRE of 3.1%, Pearson correlation 0.997, and correctly identifies 94 out of 100 busiest links on average.

5.3 Effect of Damping Factor

Without demand weighting ($\gamma = 0$), MRE increases monotonically with d (Table 3). With $\gamma = 0.20$, the model achieves Overall MRE = 0.0383 at $d = 0.96$ (Table 1)—a $5\times$ reduction in ground-truth dependence with improved accuracy.

6 Event Detection and Prediction

We validate the model against two documented real-world events: an unplanned traffic crash and a planned sporting event.

6.1 I-15 Multi-Vehicle Crash (September 20, 2018)

A four-vehicle collision blocked multiple southbound lanes of I-15 at 14400 South, Draper, from approximately 7:30–11:00 AM [2]. We compare crash-day PageRank mass on the 21 I-15 SB links against three baseline Thursdays (Sept 6, 13, 27).

Table 3: Damping factor sweep (49 timeframes, $\gamma=0$).

d	Top-100 MRE	Overall MRE	Pearson r
0.80	0.0312	0.0464	0.9969
0.84	0.0409	0.0605	0.9949
0.88	0.0558	0.0817	0.9912
0.92	0.0820	0.1177	0.9829
0.96	0.1443	0.1947	0.9565

Table 4: I-15 crash: PageRank deviation from baseline.

Time	Phase	I-15 SB	I-15 NB	Surface
06:00	Pre-crash	+1.7%	+96.5%	+10.4%
08:30	Peak	−18.9%	−92.7%	−58.6%
10:30	Rebound	+291.4%	−40.9%	−8.3%
11:30	Recovery	+36.7%	+37.1%	+15.2%

The model captures the complete crash lifecycle (Table 4): pre-crash calibration (+1.7%, confirming fair comparison), peak disruption (−18.9% SB, −92.7% NB from rubbernecking), clearance rebound (+291.4% SB as pent-up traffic releases), and recovery (+36.7%). The demand-weighted model ($d = 0.96$, $\gamma = 0.20$) produces virtually identical deviations (−18.9%, +292.4%), confirming crash detection with only 4% ground truth.

6.2 Utah vs. #10 Washington Football Game (September 15, 2018)

A near-capacity crowd of 47,445 attended the Saturday night game at Rice-Eccles Stadium (kickoff 8:00 PM) [12]. Documented road closures include 500 South lane reduction and Guardsman Way full closure [10]. We compare game day against four baseline Saturdays (Sept 1, 8, 22, 29) on 93 event-affected links.

The model detects the full game-day cycle: −85.9% on 500 South at 14:00 (road closures), +463.4% at 19:00 (arrival surge), +247.9% at stadium core at 23:00 (post-game departure).

7 The Role of μ : Per-Link Disruption Control

7.1 Why μ Is the Only Effective Control Parameter

The self-loop probability $P[i, i] = sw_i / (sw_i + \sum_k w_{ik})$ is dominated by $sw_i = \mu \cdot \ell_i / s_i$ at $\mu = 20$ (typically > 99%). We tested five types of intervention on crash-affected links:

Only μ produces meaningful changes (Table 5) because it operates on the 99.5% retained mass, while all other parameters affect only the 0.5% outgoing flow.

7.2 Crash Prediction Without Crash-Day Data

Using September 13 (normal Thursday) as input and reducing μ on the 21 I-15 SB links:

During the active diversion phase, μ -control reduces corridor error by 91–100% (Table 6). The same approach on the football game yields 94–98% error reduction on stadium-area links during closure phases.

Table 5: Effect of different interventions on I-15 SB corridor mass.

Intervention	What changes	Max effect
Speed = 0.1 m/s	Outgoing weights	−0.44%
Lanes = 1	Outgoing weights	−0.54%
Remove from adjacency	Zero outgoing	+0.67%
Zero demand weight	Zero attraction	+0.67%
$\mu = 0.01$	Self-loop: 99.5% $\rightarrow$ 9.6%	−64.4%

Table 6: I-15 SB prediction error at key crash times.

Time	$\mu=20$	Best μ	Best err	Improv.
08:00	34.1%	0.10	0.0%	+100%
08:30	179.1%	0.01	11.5%	+93.6%
08:45	380.7%	0.00	26.4%	+93.1%
10:15	142.2%	0.01	2.6%	+98.1%

7.3 Cross-Validated Baseline: Tuesday μ Profile

To confirm that $\mu = 20$ is genuinely optimal on normal days, we perform leave-one-out cross-validation across four Tuesdays (no known events). For each of 96 timeframes, we sweep μ and find the value minimising Overall MRE.

Result: The optimal μ stays in the range 10–20 across all 96 slots, with only 1% improvement over fixed $\mu = 20$. The correlation between raw ground-truth gap (between Tuesdays) and model prediction error is $r = 0.9999$ —the model adds less than 4% distortion to the input data. The model error is 96.6% of the raw input gap, confirming that the error comes from week-to-week GPS variation, not model miscalibration.

This establishes the baseline: $\mu \approx 20$ on normal days, so any significant deviation signals a real disruption.

7.4 Practical Protocol

For real-time event detection and scenario analysis:

1. Build the demand-weighted matrix ($\gamma = 0.20$) once using historical data.
2. For **detection**: run PageRank with current GPS data and compare to baseline. Deviations beyond $\pm 2\sigma$ flag anomalies.
3. For **prediction**: when a disruption is reported on specific links, set $\mu \rightarrow 0.01$ on those links and rerun PageRank with the most recent normal-day data. The output predicts the new traffic equilibrium in <10 ms on GPU.
4. **Severity mapping**: $\mu = 20$ (normal), $\mu \sim 1$ (partial restriction), $\mu \sim 0.01$ (full closure).

8 Conclusion

We presented a Two-Phase PageRank model with travel-time self-loops and demand-weighted transitions for urban traffic distribution estimation. The model achieves 3.1% Top-100 MRE on 49

held-out timeframes and detects real-world disruptions (crashes, sporting events) through PageRank deviation analysis.

Three parameters control the model:

- **Damping** d : balances matrix topology against ground truth (optimal: 0.96 with demand weighting).
- **Demand sensitivity** γ : encodes structural traffic demand into the matrix (optimal: 0.20, validated across 3 independent splits).
- **Dwell-time** μ : controls per-link mass retention—the only effective parameter for local disruption modelling (default: 20, disrupted: 0.01).

The key insight is the separation of concerns: γ handles global prediction accuracy, while μ handles local disruption response. Together, they enable a model that uses only 4% ground truth yet achieves high accuracy and can predict event-driven traffic redistribution without any event-day data.

Future work includes time-varying μ schedules matched to incident timelines, extension to multi-modal networks, and integration with real-time incident detection systems.

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